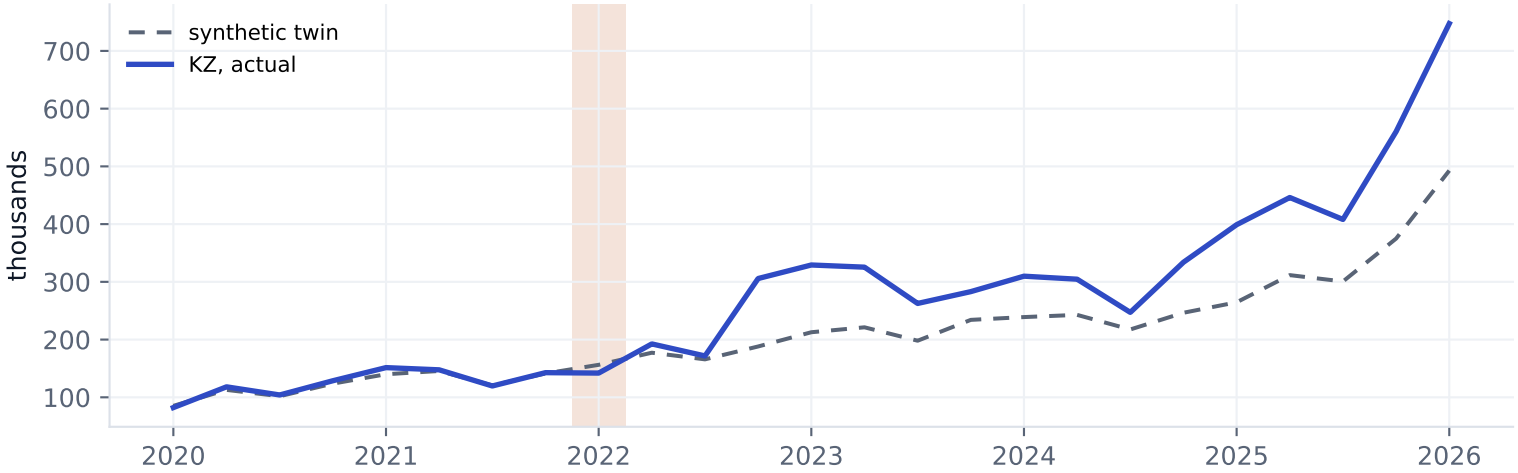
Kazakhstan, 2022

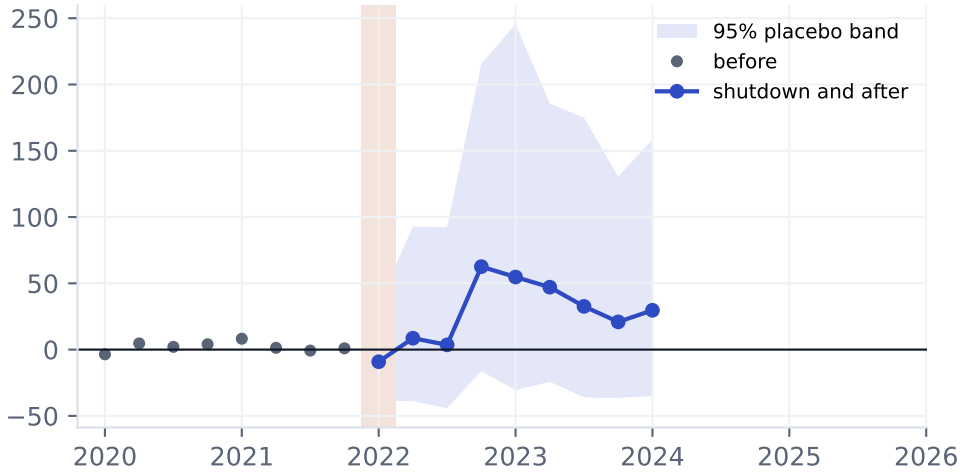
Change in git pushes attributed to KZ in 2022Q1 vs a synthetic twin (106 donor economies, 8 pre quarters)

Method	vs twin	95% interval	p normal	p perm
Synthetic DiD	-9.2%	-39% to +34%	0.630	0.393
Synthetic control	-14.7%	-46% to +33%	0.485	0.252
Augmented synthetic control	-14.3%	-41% to +25%	0.419	0.252

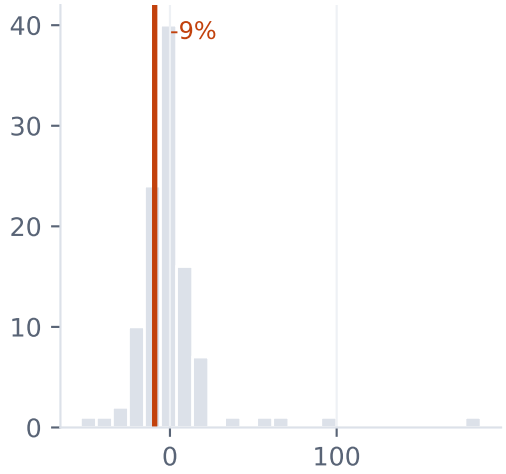
Git pushes per quarter, attributed to KZ



Gap to the twin (%)



Same design run on every donor (%)



- Conformal test. p = 0.333. With 9 quarters the smallest possible p is 0.111, so this test can't give an interval here.
- Backdating. Fitting without 2021Q3 and 2021Q4, the twin misses those quarters by -12%, -9%. Inside the placebo band.
- Breakdown. The interval already includes zero, so there is nothing to break.
- Leave one donor out. Dropping each of the 29 donors with weight over 1% keeps the estimate between -10.1% and -8.0%.
- Holm adjusted p across the nine registry events is 1.000. Does not pass every check in the analysis plan, so I don't call it a clear drop.
- What happened. Nationwide shutdown from 5 Jan 2022 to 10 Jan 2022, with short windows of access each day. Source https://pulse.internetsociety.org/en/shutdowns/kazakhstan-cuts-off-internet-access/

GitHub activity placed in the country by IP address, not national output. VPN users get placed where the VPN exits.